

EchoVision: Experiencing Bat Echolocation via Mixed Reality

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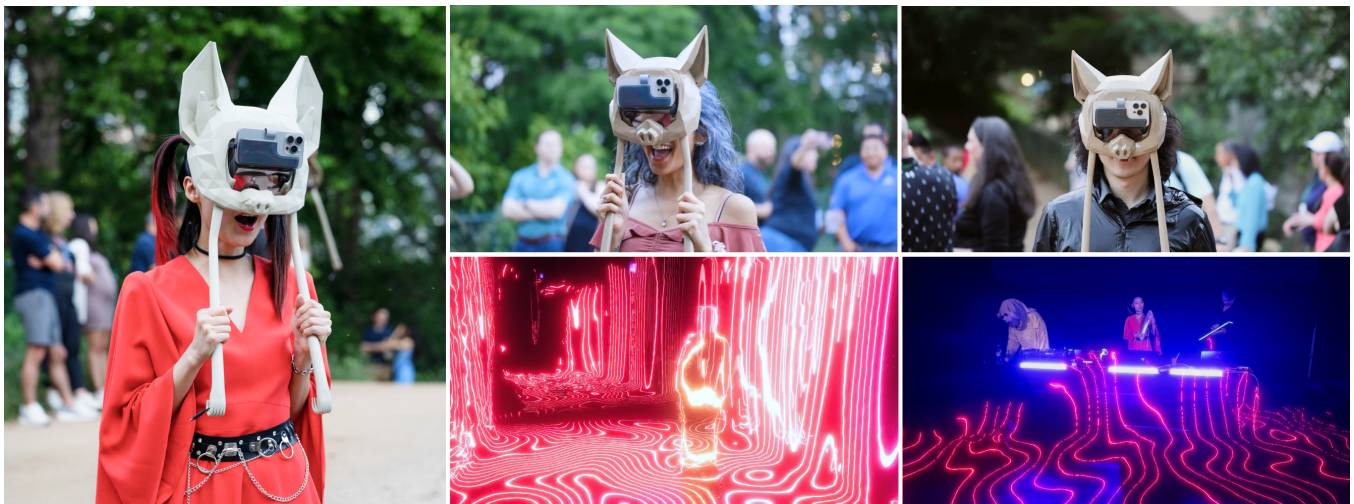


Figure 1: EchoVision is an interactive mixed reality art experience that simulates bat echolocation using custom-designed, handheld, bat-shaped masks based on the open-source HoloKit headset.

Abstract

EchoVision is an interactive mixed reality art experience that simulates bat echolocation using custom-designed, handheld, bat-shaped masks based on the open-source HoloKit headset. As users initiate sounds, they experience echolocation—the remarkable navigation method bats employ in darkness—through visual representations of propagating echoes mapped onto the real-world 3D shape of the surrounding environment, scanned in real-time by the LiDAR sensor of a smartphone mounted on the masks. Unlike traditional indoor mixed reality headsets that often require time-consuming

per-person setup and calibration, EchoVision’s ergonomic handlebar design allows for easy passing and sharing among numerous participants, enabling quick engagement with the experience. This feature is especially valuable for in-the-wild, large-scale, pop-up exhibitions in bat habitats, such as Austin’s Congress Avenue Bridge, enhancing high-traffic scientific education through deeper empathetic engagement. By providing a unique glimpse into how non-human creatures perceive their sensory world, EchoVision promotes an ecocentric design perspective, fostering interspecies understanding and appreciation.

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CCS Concepts

• **Human-centered computing** → Collaborative and social computing theory, concepts and paradigms; Visualization design and evaluation methods; Interaction design theory, concepts and paradigms; Mixed / augmented reality.

Keywords

Augmented Reality, Mixed Reality, Handheld Mixed Reality Mask, Ecocentric design, Non-Anthropocentric design

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Bats often face unwarranted fear and misconceptions. Many erroneously believe bats are blind, disease-ridden creatures that seek to entangle themselves in human hair or suck blood. These myths, stemming from folklore, media exaggeration, and misinformation, have been exacerbated by the perception of bats as primary disease carriers, especially during the COVID-19 pandemic [MacFarlane and Rocha 2020]. In reality, bats are diverse, mostly harmless mammals crucial to ecosystems. They serve as pollinators, seed dispersers, and insect controllers, helping maintain ecosystem balance and biodiversity [Platto et al. 2021]. Despite their ecological importance, bats suffer habitat loss and persecution. Public education about bats' true nature is crucial for their conservation.

Thomas Nagel's famous philosophical inquiry "What Is It Like to Be a Bat?" [Nagel 1974] highlighted the challenge of understanding the subjective experience of beings with fundamentally different sensory systems, using bats and their echolocation abilities as a prime example. Inspired by this very concept, we created EchoVision, an interactive mixed reality art experience that allows users to experience bat echolocation, using custom-designed, handheld, bat-shaped masks based on the open-source HoloKit headset [Hu et al. 2024]. As users initiate sounds, the system visualizes sound echoes overlaid on reality, propagating through a real-time 3D reconstruction of the surrounding environment, captured by a LiDAR scanner on a smartphone mounted on the mask. EchoVision leverages the similarity between bat echolocation and LiDAR scanning technology: Bats emit high-frequency sound waves and interpret the returning echoes, while LiDAR devices use laser pulses and measure their reflections. As shown in Figure 3, EchoVision produces an artistic simulation of echolocation. The visual effects dynamically respond to the volume and pitch of the user's voice input, as well as the contours of the surrounding environment. This immersive experience allows humans to glimpse into a bat's perceptive world, offering participants a unique, empathetic perspective of a bat's reality.

We premiered EchoVision in a natural setting—the bat habitat under the Congress Avenue Bridge (the "Bat Bridge") in Austin, Texas, as shown in Figure 2. This popular tourist spot, where millions of Brazilian free-tailed bats forage after sunset, provided an ideal location to engage a large, interested audience. Unlike traditional indoor mixed or virtual reality exhibitions that often require expensive equipment and time-consuming per-person setup [Arvola et al. 2021], EchoVision offers a more inexpensive, self-propagating, and setup-free solution, thanks to HoloKit's cost-effectiveness and accessibility. The mask's ergonomic handlebar design allows for easy passing and sharing among participants, enabling quick engagement. Its bat-shaped appearance serves as an inviting curiosity for uninformed passersby. During this pop-up exhibition, over 200 attendees—including registered members and casual passersby—listened to a lecture and experienced EchoVision beneath the "Bat Bridge." This high-traffic event encouraged large-scale scientific education and empathetic engagement, helping to



Figure 2: (a)(b) Over 200 attendees, including registered participants and casual passersby, experienced EchoVision beneath the "Bat Bridge"—the Congress Avenue Bridge, which is home to millions of Brazilian free-tailed bats. (c) People gathered closer to the bridge to watch the bats emerge in search of food after sunset. (d)(e) The mask handlebars are designed for easy passing to nearby people.

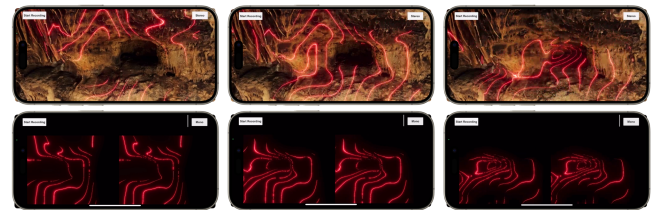


Figure 3: (Top Row) Monoscopic view of EchoVision on the phone screen, visualizing sound propagation in the cave; (Bottom Row) Stereoscopic rendering of EchoVision for HoloKit.

dispel myths, and inspire a deeper connection to and respect for bats.

EchoVision transforms misconceptions into appreciation by highlighting the profound differences in how various species interact with their environments and emphasizing the vital ecological contributions of these often misunderstood creatures. By adopting an ecocentric design approach, EchoVision celebrates the diverse ways non-human creatures interact with their environment, fostering a broader, more-than-human understanding of our shared ecosystem.

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